

FedHealth-XAI: A Robust Federated Learning Framework for Privacy-Preserving and Interpretable Clinical Diagnostics

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FedHealth-XAI: A Robust Federated Learning Framework for Privacy-Preserving and Interpretable Clinical Diagnostics

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Abstract—

The problem of training machine learning models within the healthcare domain is, at its core, a problem of access — specifically, the absence of it. Patient records held by one institution are, in most legal jurisdictions, inaccessible to any external party, including researchers developing diagnostic tools that could benefit future patients at that very institution. Regulatory instruments such as the Health Insurance Portability and Accountability Act (HIPAA) codify this restriction, prohibiting the centralization of sensitive patient data across organizational boundaries. The consequence is that conventional training pipelines, which assume the unrestricted availability of pooled data, are rendered legally impractical in the majority of real-world clinical scenarios.

A secondary complication arises from the nature of the data itself. Even were legal barriers absent, medical records collected across geographically and institutionally distinct sites exhibit pronounced distributional heterogeneity. Variations in patient demographics, local disease burden, and prevailing clinical practice patterns produce datasets that are, in statistical terminology, non-independent and identically distributed (non-IID). This heterogeneity introduces instability into the training of distributed models, degrading both convergence behavior and the generalizability of the learned representations. Alongside these challenges, the opacity of modern deep learning architectures further limits clinical adoption: practitioners asked to act on the predictions of a model they cannot interpret will, not unreasonably, decline to do so.

The FedHealth-XAI framework presented in this paper represents a concerted effort to address all three of these constraints within a single, coherent pipeline. Implemented in PyTorch and coordinated through the Flower federated learning library, the system enables five simulated institutional clients to train a shared diagnostic model without transmitting raw patient records. Each site handles its own data preparation independently—using median imputation for missing values and normalizing features locally—without sharing any statistical details across institutions. The training process is guided by the FedProx optimization algorithm, which adds a proximal regularization term to keep local updates aligned with the evolving global model, helping to address convergence issues caused by non-IID data.

Formal privacy protections are ensured through Meta’s Opacus library, which applies Differentially Private Stochastic Gradient Descent (DP-SGD). Gradient clipping and calibrated Gaussian noise injection together ensure that the information carried in transmitted updates cannot be exploited to reconstruct individual patient records. Model interpretability is addressed through SHAP (SHapley Additive exPlanations), applied post-hoc to the final global model; the resulting feature attribution scores give clinicians a principled basis for evaluating whether the model’s reasoning aligns with established medical knowledge. Evaluated on the UCI Heart Disease Dataset, FedHealth-XAI achieves an AUC-ROC of 0.91 and an F1-score of 0.73 — performance levels that indicate clinical-grade utility can be sustained even under the combined constraints of decentralization, privacy noise, and data heterogeneity.

Index Terms—federated learning, differential privacy, explainable AI, SHAP, FedProx, healthcare, clinical diagnostics, non-IID data

1. INTRODUCTION

1.1 The Clinical Deep Learning Dilemma

Among the various subfields where deep learning has registered meaningful empirical progress, clinical diagnostics stands out for the breadth and consistency of its gains. The capacity of these models to identify pathological patterns in medical images, to extract prognostic signals from longitudinal laboratory data, and to integrate heterogeneous inputs from electronic health records has, over the past decade, produced systems whose predictive accuracy rivals or exceeds that of trained specialists on narrowly defined tasks. Yet the gap between laboratory benchmark performance and deployed clinical utility remains wide, and the explanation for that gap is, to a considerable degree, a data problem.

Deep learning models of the kind that achieve state-of-the-art results on diagnostic benchmarks are data-hungry in a very particular way: they require not merely large quantities of examples, but examples that capture the full breadth of variation present in the clinical population the model is intended to serve. A model trained on data from a single institution — even a large and well-resourced one — may perform poorly when deployed elsewhere, because the patient population, imaging equipment, documentation conventions, and disease spectrum at the deployment site differ systematically from those represented in the training corpus. Generalizability, in other words, requires diversity, and diversity requires data from many institutions.

The difficulty is that assembling such multi-institutional training sets is far from straightforward. Data protection legislation in most jurisdictions — the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) across the European Union being the most prominent examples — imposes stringent restrictions on the transfer of identifiable patient information across organizational boundaries. Individual institutions have responded by treating their data holdings as internal assets, maintaining them in isolated repositories that are effectively inaccessible to external collaborators. This produces a situation in which the aggregate diagnostic value of distributed clinical datasets is largely unrealized: each institution's records remain siloed, training data for AI systems remains narrow, and model generalizability suffers as a consequence.

Some attempts have been made to navigate this tension through formal data-sharing agreements, federated research cohorts, or de-identification protocols that reduce but do not eliminate privacy risk. These approaches have demonstrated that limited cross-institutional collaboration is possible but have not resolved the underlying structural problem: the healthcare ecosystem as a whole continues to lack an efficient, privacy-compliant mechanism for training machine learning models on the distributed data that would make them clinically reliable. The gap between what deep learning could deliver with adequate data access and what it currently delivers within real regulatory constraints remains a defining challenge for the field.

1.2 Challenges in Standard Federated Learning

Federated learning emerged as a principled response to precisely this impasse. Rather than attempting to move data to a central model, the federated paradigm inverts the relationship: the model is distributed to the data. Participating institutions each train a local copy of the model on their own patient records and periodically share only the resulting parameter updates with a central aggregation server. The server combines these updates — typically through some form of weighted averaging — to produce a refined global model, which is then redistributed for the next round of local training. Raw data never leaves the institution that holds it; only model updates are transmitted. McMahan and colleagues' Federated Averaging (FedAvg) algorithm, introduced in 2017, provided the initial formal demonstration that this approach was technically feasible and ignited a substantial body of subsequent research.

The practical limitations of standard federated approaches, however, become apparent quickly when they are applied to real healthcare data. The most fundamental of these is the non-IID problem. In federated learning theory, convergence guarantees typically rest on the assumption that local datasets are drawn from the same underlying distribution. Clinical data violates this assumption in virtually every realistic

deployment scenario. An urban academic medical center may serve a demographically diverse population presenting with a wide range of acute and chronic conditions across many organ systems. A rural primary care clinic, by contrast, may see a predominantly elderly population with a narrower spectrum of chronic conditions and far lower rates of acute presentations. The statistical distance between the local distributions at these two institutions can be substantial, and when each trains a local model that reflects its own population, the resulting updates may be poorly aligned — or even contradictory. FedAvg averages them regardless, producing a global model that may converge slowly, converge to a suboptimal solution, or fail to converge at all.

This phenomenon, often referred to as *client drift*, represents a qualitative failure mode rather than merely a quantitative performance penalty. It is not simply that the global model is somewhat less accurate than it would be with IID data; in severe cases, the global model may perform worse than any of the local models would if trained independently. The problem is well-documented in the federated learning literature and has motivated considerable algorithmic work aimed at stabilizing optimization in the presence of data heterogeneity.

A separate limitation concerns the privacy properties of the federated paradigm itself. Standard federated learning is often described as privacy-preserving, and in a limited sense this description is accurate: patient records remain on local servers and are never transmitted. What is transmitted, however, are gradient updates or model weight deltas, and it has been shown — most influentially by Shokri et al. (2017) and Zhao et al. (2020) — that these updates can leak substantial information about the training data from which they were derived. Model inversion attacks can approximately reconstruct training samples from gradient information; membership inference attacks can determine with non-trivial accuracy whether a specific individual's data contributed to a given model's training. The privacy guarantee offered by standard federated learning is therefore practical rather than formal: it reduces the surface area for data exposure but does not eliminate it. For applications involving sensitive patient information, this distinction is not academic — it is the difference between a system that meets regulatory privacy requirements and one that does not.

1.3 The Need for Algorithmic Transparency

Suppose, for the sake of argument, that the privacy and convergence problems described above were fully resolved — that a federated algorithm existed which converged reliably on non-IID data and provided formal, mathematically rigorous privacy guarantees for all transmitted information. Would the resulting system be deployable in clinical practice? Not necessarily, for a reason that has nothing to do with privacy or statistical optimization: interpretability.

Clinical decision-making is not simply a matter of obtaining a probability estimate and acting on it. Medical professionals carry both legal and ethical responsibility for their decisions. They are trained to assess evidence, weigh different possibilities, and explain their reasoning to patients, colleagues, and, when required, to courts or regulatory authorities. A model that produces a diagnosis or risk score without any accompanying explanation of how that output was derived places the clinician in a professionally untenable position. They are being asked to delegate part of a consequential decision to a system whose reasoning they cannot inspect, cannot challenge, and cannot explain to others. Most clinicians will, quite rationally, decline.

The standard response to concerns about AI transparency is to point to aggregate performance metrics: if the model achieves high AUC and F1-score on a held-out test set, should that not be sufficient evidence of its reliability? The answer, for most clinical applications, is that it is not. Aggregate metrics characterize model behavior across a population of test cases; they say nothing about whether the model's reasoning for any particular case is medically sound. A model might achieve high overall accuracy by exploiting a spurious correlation present in the training data — one that holds in the training population but fails in

deployment — and aggregate metrics would not reveal this until after the deployment had caused harm. Feature-level explanations, by contrast, allow clinicians to verify that the model is attending to the right clinical signals for each prediction, providing a basis for the kind of qualitative validation that complements quantitative benchmarking.

Explainability is not, therefore, a cosmetic concern that can be addressed after the core technical problems are solved. It is a prerequisite for clinical adoption, and in many regulatory contexts it may also be a legal requirement. Building it into the framework from the outset — rather than attempting to retrofit it after the fact — is both technically and practically the more defensible approach.

1.4 Formal Privacy Guarantees through Differential Privacy

While federated learning reduces the need for direct data sharing, it does not, in isolation, provide rigorous protection against information leakage through model updates. Recent work has demonstrated that gradient exchanges can inadvertently encode sensitive details about individual training samples, rendering naive federated implementations insufficient for high-stakes domains such as healthcare. This limitation necessitates the incorporation of formal privacy-preserving mechanisms that go beyond structural decentralization.

Differential Privacy (DP) provides a principled way to reason about privacy in such settings. Rather than depending on assumptions about limited exposure, it offers a formal guarantee: the inclusion or exclusion of any single data point won't noticeably change the model's output. In practice, this is usually done with Differentially Private Stochastic Gradient Descent (DP-SGD), where gradients are clipped to limit sensitivity and then adjusted with carefully calibrated noise. These steps may appear simple, but together they fundamentally alter what can be inferred from the training process.

What makes DP-SGD stand out in federated healthcare systems isn't just its strong theoretical roots, but the fact that it actually works in practice: it can be integrated into existing deep learning pipelines without requiring major architectural changes, which is important in real deployments where systems are already complex. At the same time, privacy accounting methods such as Rényi Differential Privacy allow tracking how much privacy "budget" is being consumed over time. This introduces a more transparent way of reasoning about privacy, although it also forces developers to confront an unavoidable trade-off between privacy strength and model performance.

For this reason, differential privacy is better understood not as an add-on feature, but as a necessary design consideration in sensitive domains. Especially in healthcare, where even indirect data leakage can have serious consequences, relying solely on decentralization is insufficient. Incorporating DP mechanisms shifts the system toward verifiable privacy guarantees, even if it requires careful tuning and acceptance of some performance overhead.

2. RELATED WORK

The design of FedHealth-XAI draws on four distinct but interrelated bodies of research: federated learning architectures, methods for handling statistical heterogeneity in distributed training, cryptographic privacy techniques for deep learning systems, and explainable AI methods applicable to clinical decision support. A brief survey of each domain situates the contributions of the present work within the existing literature.

2.1 Federated Learning in Healthcare

The concept of federated learning was introduced by McMahan et al. in 2017, whose Federated Averaging algorithm demonstrated, for the first time, that a single global model could be trained collaboratively

across a large number of distributed clients without centralizing their data. The technical elegance of the approach attracted immediate attention, and the healthcare community in particular recognized its potential applicability to the longstanding problem of multi-institutional data access. Within a year, researchers including Brisimi et al. (2018) had begun adapting federated methods to the specific demands of medical data, demonstrating that meaningful predictive models could be trained on EHRs distributed across hospital systems while keeping patient records local.

These early demonstrations were important. They established feasibility, provided proof-of-concept implementations, and helped frame federated learning as a credible response to healthcare's data-sharing problem. At the same time, they shared a common and consequential simplifying assumption: that the data distributions across participating institutions were roughly similar — Independent and Identically Distributed in the statistical sense. In practice, this assumption almost never holds. Patient populations vary substantially across hospitals, and that variation has deep roots: the socioeconomic composition of the catchment area shapes who presents for care and with what conditions; regional epidemiological patterns determine disease prevalence; differences in specialty focus, referral patterns, and institutional protocols influence what gets documented and how. A hospital that treats a disproportionate share of elderly patients will have a very different distribution of comorbidities, drug regimens, and laboratory values than one serving a younger, more transient urban population.

The consequences of applying standard FedAvg to such genuinely non-IID distributions have been documented in some detail. Models can exhibit what is variously described as client drift, gradient divergence, or local overfitting: the local model at each site converges toward a solution that is well-adapted to that site's data but poorly aligned with the global optimum. When these divergent local updates are averaged, the resulting global model may represent none of the participating sites well. Performance degradation relative to centralized training can be severe, and in some experimental settings, federated models trained on non-IID data have been shown to perform worse than any individual local model. These findings highlight a fundamental gap between the theoretical guarantees that federated optimization theory provides under IID conditions and the empirical performance the approach delivers on real clinical data.

2.2 Mitigating Statistical Heterogeneity

The recognition that data heterogeneity represents a serious practical obstacle to federated learning prompted a line of research focused specifically on stabilizing optimization under non-IID conditions. The most widely adopted result of this effort is the FedProx framework of Li et al. (2020), which modifies the local training objective by adding a proximal regularization term. The term penalizes the distance between the current local parameters and the global model parameters broadcast at the beginning of each round, effectively preventing local models from drifting too far from the global solution during the local training phase. Convergence guarantees that hold under data heterogeneity can be established for FedProx under conditions where FedAvg provides no such guarantees.

FedHealth-XAI adopts FedProx as its optimization backbone, with the proximal coefficient set to $\mu = 0.01$ following evaluation of convergence behavior across the five simulated institutional nodes. This is complemented by a preprocessing protocol that enforces strict data locality: all imputation and normalization operations are carried out independently at each node, without any sharing of summary statistics across sites. This combination addresses a dimension of the heterogeneity problem that most FedProx implementations in the literature have not explicitly considered — the possibility that preprocessing, if performed using shared global statistics, could itself introduce a form of cross-institutional data leakage.

2.3 Differential Privacy in Deep Learning

The federated learning paradigm addresses the problem of raw data exposure by keeping patient records local; it does not, by itself, address the problem of information leakage through model updates. The vulnerability is real and has been demonstrated experimentally. Carlini et al. (2019) showed that language models trained on sensitive text could be induced to reproduce verbatim memorized training sequences, indicating that gradient information can carry detailed traces of individual training examples. In the federated context, Shokri et al. (2017) demonstrated that shared gradient updates could be exploited to infer properties of individual training records, and Zhao et al. (2020) showed that in some settings, training data could be approximately reconstructed from gradients with high fidelity. These results establish that gradient sharing, without additional safeguards, provides privacy in appearance rather than in substance.

The principled solution is differential privacy, whose application to deep learning was formalized by Abadi et al. (2016) through the DP-SGD algorithm. The approach proceeds in two steps: first, per-sample gradients are clipped to bound the maximum influence of any single training example on the update; second, Gaussian noise calibrated to the clipping threshold is added to the aggregate gradient before it is applied. The clipping bound and noise magnitude together determine the privacy budget ϵ , which quantifies the worst-case privacy loss per training step and can be accumulated across training rounds through formal accounting methods. A smaller ϵ corresponds to stronger privacy but, generally, to a larger degradation in model utility.

FedHealth-XAI implements DP-SGD through Meta's Opacus library, which provides efficient per-sample gradient computation within the standard PyTorch training loop. The noise multiplier is set to 0.1 and the gradient clipping threshold to 1.0; these values were selected to maintain model performance above the threshold of clinical utility while ensuring a privacy budget that falls within operationally defensible bounds. At every communication round, privacy expenditure is monitored and recorded using Rényi differential privacy accounting, giving ongoing insight into the balance between privacy and utility as training advances.

2.4 Explainable AI for Clinical Decision Support

The interpretability problem in clinical AI has a long history that predates deep learning. Even earlier generations of machine learning methods — support vector machines, ensemble methods, neural networks of the type common in the 1990s — attracted criticism for their opacity, and various post-hoc explanation methods were proposed as remedies. The problem has, if anything, become more acute as deep learning architectures have grown more complex: a model with millions of parameters, multiple non-linear layers, and attention mechanisms has an internal structure that is not amenable to inspection by conventional means.

Lundberg and Lee (2017) proposed SHAP as a principled solution grounded in cooperative game theory. The key insight is to treat the features of an input as players in a coalition game, where the “payoff” is the model's prediction, and to compute the Shapley value — a concept from game theory with a number of desirable fairness and consistency properties — for each feature. The resulting attribution scores are locally accurate (they sum to the difference between the model output and a baseline), consistent across the feature set, and applicable to any differentiable model through the DeepExplainer variant used here. FedHealth-XAI applies SHAP DeepExplainer to the final aggregated global model, demonstrating that feature attributions of clinical significance — including the risk contributions of variables such as age and sex — remain stable and interpretable even after the privacy noise introduced by DP-SGD and the distributional averaging inherent in federated aggregation.

2.5 Summary of Contributions vs. Existing Literature

A survey of the literature across these four domains reveals a consistent pattern: existing work tends to address one or at most two of the challenges that confront clinical federated AI. Papers that combine

federated learning with privacy mechanisms typically evaluate on IID or near-IID data distributions and pay limited attention to the non-IID convergence problem. Works focused on handling data heterogeneity rarely incorporate formal differential privacy, treating the gradient-sharing problem as outside their scope. Explainability research in healthcare has, with few exceptions, been conducted in centralized settings where the full training dataset and model are available for explanation; the question of whether explanation methods survive the perturbations introduced by federated aggregation and differential privacy has received little attention. Table I summarizes these gaps and situates FedHealth-XAI as a framework that addresses all three dimensions simultaneously rather than in isolation.

TABLE I
Comparison of FedHealth-XAI With Existing Approaches

Author (Year)	Core Framework	Primary Contribution	Limitation Addressed by FedHealth-XAI
McMahan et al. (2017)	FedAvg	Pioneered decentralized model training via weight aggregation	Fails to converge on heavily skewed, Non-IID clinical data
Li et al. (2020)	FedProx	Introduced proximal term (μ) to stabilize Non-IID training	Lacked cryptographic privacy (DP) and interpretability (XAI)
Abadi et al. (2016)	DP-SGD	Established mathematical foundations for DL privacy	High noise degrades utility; rarely applied in multi-node FL
Lundberg & Lee (2017)	SHAP	Game-theoretic feature importance for model transparency	Traditionally centralized; FedHealth-XAI proves XAI survives DP-SGD
FedHealth-XAI (Ours)	FedProx + DP + XAI	End-to-end: localization, DP, FedProx, and SHAP	Solves the complete clinical triad: Heterogeneity, Cryptography, and Transparency

3. METHODOLOGY

3.1 The “Local-First” Preprocessing Pipeline

Compliance with clinical data privacy requirements begins before model training — at the preprocessing stage. A straightforward but consequential design choice in FedHealth-XAI is the complete localization of all data preparation operations. Global normalization, in which scaling parameters are computed from data pooled across participating sites, is explicitly avoided: such parameters encode distributional information about the aggregate dataset, and sharing them would constitute a form of cross-institutional data leakage, however subtle. Instead, each of the five simulated hospital nodes independently manages its own preprocessing pipeline, with no communication with other nodes at this stage.

Missing values within each local dataset are addressed through scikit-learn’s SimpleImputer, configured for median imputation. The choice of median over mean is deliberate and clinically motivated. Healthcare datasets routinely contain values that are extreme by statistical standards but entirely legitimate clinically — a patient presenting with a cholesterol of 550 mg/dL or a resting blood pressure of 190 mmHg is not an outlier to be discounted but a case of potential clinical significance. Mean imputation would allow such

values to distort the imputed values for missing entries at the same site; the median is robust to this class of influence and produces imputed values more representative of the typical clinical presentation.

Feature standardization is applied through site-specific StandardScaler instances, each fitted exclusively to the local data distribution. The features are mapped to Z-scores using only the mean and variance computed from the records at that institution. No site shares its scaling parameters with any other. A secondary consequence of this approach — beyond the privacy benefit — is that the global model is trained against inputs normalized according to locally varying standards, which may better reflect the kind of distributional variation it will encounter if deployed across heterogeneous clinical settings.

3.2 Algorithmic Stability: The FedProx Regularizer

Client drift, the tendency of local models to move away from the global optimum when trained on non-IID local data, is addressed through the FedProx optimization framework. At each communication round, the local training objective at node k is adjusted with a proximal penalty term, resulting in the modified loss function:

$$h_k(w) = F_k(w) + \frac{\mu}{2} \|w - w^t\|^2(1).$$

Here, $F_k(w)$ represents the empirical loss at client k , w^t is the global model broadcast at the start of round t , and μ controls the strength of the constraint. This penalty discourages local solutions from straying too far from the global model, reducing the risk of any single institution overfitting to its own patient population and thereby distorting the aggregated outcome. The coefficient $\mu = 0.01$ was selected empirically as providing adequate regularization without unduly restricting local learning; increasing μ beyond this value was found to slow convergence without meaningfully improving the stability of the global model.

3.3 Cryptographic Layer: DP-SGD via Opacus Differentially Private Stochastic Gradient Descent is implemented using Meta's Opacus library, integrated into the PyTorch training loop at each client node. The privacy mechanism unfolds in three sequential steps at every training iteration.

- **Gradient Clipping:** The L_2 norm of each individual gradient is clipped to a maximum of 1.0 before mini-batch aggregation. This ensures that no single patient record can disproportionately influence the update, forming the first safeguard against recovering individual-level information.
- **Gaussian Noise Injection:** Once clipped, the aggregated gradient is perturbed with Gaussian noise, scaled at 0.1 relative to the clipping threshold, before being sent to the central Flower server. Together, clipping and noise injection enforce the DP-SGD mechanism, delivering formal privacy guarantees quantified by the privacy budget parameter ϵ .
- **Privacy Budget Tracking:** At each communication round, cumulative privacy expenditure is tracked using Rényi differential privacy accounting, which provides tighter bounds on privacy loss compared to classical composition methods.

The tracked budget enables continuous evaluation of the privacy-utility trade-off across training rounds and provides a principled basis for verifying that the final model operates within thresholds consistent with HIPAA-aligned data protection standards.

3.4 Orchestration and Execution Workflow

The federated training process is coordinated through Flower's Virtual Client Engine (VCE), with Ray handling parallel execution across the five simulated hospital nodes. At the commencement of each of the five communication rounds, the central server distributes the current global model weights to all participating clients. Each client trains locally for five epochs on its partitioned dataset, using mini-batches

of size 16. The optimization combines DP-SGD with the FedProx-augmented loss function. Following local training, the updated parameter vectors¹ — having been subjected to per-sample gradient clipping and Gaussian noise injection — are returned to the server. The server performs weighted aggregation of the received updates, with weights proportional to each client's local dataset size, to produce the updated global model for the next round.

3.5 Interpretability Layer: Post-Hoc XAI After federated training is complete, the final global model is examined using the SHAP DeepExplainer framework to provide interpretability. DeepExplainer computes approximate Shapley values for each input feature by propagating explanation signals through the network using a background dataset as the reference distribution. The resulting attribution scores represent the marginal contribution of each clinical variable — Age, Sex, Chest Pain Type, and others — to each individual model prediction, with positive values indicating features that increase the predicted risk of disease and negative values indicating features that reduce it.

"The SHAP summary plot combines individual attributions across the evaluation dataset, producing a ranked view of feature importance that clinical practitioners can easily interpret. As an illustrative example, the framework might indicate that a patient's age contributed +0.4 to the predicted heart disease classification, while an elevated maximum heart rate contributed a negative attribution — shifting the prediction toward a lower-risk assessment. This type of feature-level output supports the kind of qualitative model validation that complements quantitative benchmarking: clinicians can assess whether the features weighted most heavily by the model are those that clinical knowledge would identify as most diagnostically relevant.

4. EXPERIMENTS AND RESULTS

4.1 Experimental Setup and Hyperparameters

FedHealth-XAI was evaluated in a simulated federated environment comprising five hospital nodes, each operating as an isolated data silo. The UCI Heart Disease Dataset was split into five distinct subsets, with each assigned to a separate node. This setup ensured that no data was shared between sites during either training or evaluation. Federated training was conducted over five communication rounds; within each round, every client performed five epochs of local training using mini-batches of size 16. The FedProx proximal coefficient was set to $\mu = 0.01$, with Opacus configured for a gradient clipping threshold of 1.0 and a noise multiplier of 0.1. These hyperparameters were chosen to strike a balance between convergence stability, privacy protection, and predictive performance.

4.2 Performance Metrics and Robustness:

Despite the challenges of fully decentralized training, site-specific preprocessing, and the addition of differential privacy noise, the global model achieved clinically meaningful performance. Peak accuracy reached about 80%, with an F1-score of 0.73 and an AUC-ROC of 0.91. The receiver operating characteristic curve presented in Figure 1 illustrates that this performance reflects a high and stable true-positive rate across a wide range of decision thresholds, achieved with a correspondingly low false-positive rate — a combination of particular importance in cardiac screening, where the cost of a missed positive diagnosis substantially outweighs the cost of an unnecessary follow-up.

4.3 Diagnostic Reliability and Error Analysis

Examination of the confusion matrix presented in Figure 2 reveals a specific and clinically interpretable pattern of errors. The model correctly classified 158 healthy patients and 86 with disease, while producing only 6 false positive predictions. This conservatism — high specificity relative to sensitivity — is a recognizable consequence of the differential privacy mechanism: noise injection attenuates gradient

signals and biases the model toward restraint in positive predictions. In cardiac screening applications, this pattern has practical advantages. False positives trigger clinical workups that are costly, time-consuming, and anxiety-inducing for patients; a model that maintains a very low false-positive rate at the cost of some sensitivity is, in many deployment contexts, a more acceptable tool than one that generates frequent false alarms.

4.4 Interpretability and Medical Validation

SHAP analysis conducted on the final aggregated global model, illustrated in Figure 3, identifies Age and Sex as the dominant predictors of heart disease risk in the model’s learned representation. This finding has immediate face validity: advancing age and biological sex are among the most robustly established cardiovascular risk factors in the epidemiological literature, and their prominence in the SHAP attribution scores provides a meaningful basis for trusting that the model has learned clinically relevant relationships rather than spurious correlations.

Notably, the structured clustering patterns in the SHAP interaction values — which reflect non-trivial learned relationships between features rather than simple additive effects — remained coherent following federated aggregation across five heterogeneous data silos and the injection of privacy noise throughout training. This finding addresses a concern that motivated part of the framework’s design: that differential privacy noise, by reducing the signal-to-noise ratio in gradient updates, might wash out the feature relationships captured by the model to the point where SHAP attributions became clinically uninformative. The results indicate that, at the noise levels used here, this degradation does not occur and that the interpretability layer functions as intended even in the presence of formal privacy constraints.

5. CONCLUSION AND FUTURE WORK

5.1 Conclusion

The central claim of this paper is that the three most significant obstacles to the clinical deployment of federated AI — statistical heterogeneity across participating institutions, the inadequacy of standard federated learning as a privacy mechanism, and the interpretability demands of clinical practice — can be addressed simultaneously within a single framework. FedHealth-XAI provides empirical evidence in support of this claim. By integrating FedProx-based optimization, DP-SGD via Opacus, and SHAP-based post-hoc explanation, the system achieves a clinical-grade AUC of 0.91 on the UCI Heart Disease Dataset while maintaining a tracked and bounded privacy budget and producing feature attributions that are both locally accurate and clinically coherent.

Prior work has tended to treat these three problems as separate engineering concerns, each addressed in isolation. The assumption underlying much of the federated learning literature is that privacy and convergence are the primary technical problems, with interpretability to be handled separately if at all. FedHealth-XAI challenges this framing: the results demonstrate that SHAP attributions survive both federated aggregation and differential privacy noise injection, that FedProx regularization stabilizes convergence without preventing local learning, and that DP-SGD can be parameterized to preserve clinical utility while meeting formal privacy requirements. The three objectives are not, as is sometimes assumed, fundamentally in tension; they can be balanced through principled design choices, and the resulting system represents a viable direction for responsible AI deployment in healthcare.

5.2 Future Work

Multi-Dataset Benchmarking: The present evaluation is limited to a single dataset under simulated partitioning. Validation on larger and clinically diverse datasets — including the Breast Cancer Wisconsin dataset, the Pima Indians Diabetes dataset, and ideally real EHR data collected under appropriate data use

agreements — would provide stronger evidence of generalizability across disease types, imaging modalities, and institutional contexts.

Advanced Privacy Mechanisms: Differential privacy as implemented here protects against gradient-level attacks from external adversaries but does not address the threat model in which the aggregation server itself is compromised. Future work could explore the integration of Secure Multi-Party Computation (SMPC) or Homomorphic Encryption to provide cryptographic protection at the aggregation stage, complementing the differential privacy guarantees applied during local training.

Active Learning for Imbalanced Clinical Data: The current framework treats all local training samples uniformly, without regard to their clinical informativeness or representativeness. Incorporating active learning strategies that selectively prioritize rare or high-risk cases — underrepresented minorities in clinical datasets that are often of disproportionate diagnostic importance — could improve model sensitivity to critical subpopulations without requiring additional data collection or compromising privacy guarantees.

Mobile-Edge Deployment: Extending FedHealth-XAI to operate on point-of-care devices and rural health facilities through Flower’s mobile-edge capabilities would substantially expand the practical reach of the framework, particularly in resource-constrained settings where centralized computing infrastructure is unavailable and the need for robust, privacy-preserving diagnostic support is arguably most acute.

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